Rust In Peace

Hard disks might soon be dead. But what will replace them? Here's a look at seven technologies that could be called "hard disk killers"

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Storage researchers have their work cut out all nice and neat: "all" that's needed is a way to make 1s and 0s. A Yes and a No; a hole and a bump; a groove and a ridge. This means a lot of freedom to experiment—there are a lot of materials and mechanisms researchers can tinker around with, with the goal of finding what materials can be in two states (or be brought into two states), and how to go about making those state changes. Given a decent lab, you could probably think up a storage mechanism yourself—with the caveat that what's needed is higher densities!

When you look at what's being researched and actually think about our current storage devices (and those in the past, such as punch cards), you'll see that they're primitive. They almost seem stupid. Holes in paper—how straightforward can you get? The hard disk, too, is based on an extremely simple, almost simplistic principle—different magnetic orientations on a layer of iron oxide.

It turns out we can now do much better. From a fairly lofty perspective, therefore, the hard disk is dead. This century belongs to nanotech, the science of the extremely small. And small is, of course, good news when it comes to storage, because it means higher densities. In what follows, we discuss several storage techniques in various stages of research and implementation, and with very different theoretical caps—from the one CD per square centimetre achieved using a type of rubber stamp, to the beyond-the-imagination one crore CDs on a square inch that might be achieved with the help of carbon nanotubes. Some of these technologies you'll see in the next couple of years; some you'll see sometime in your lifetime. Some will die because of competing technologies, some will tame the world's information. But they all demonstrate that innovation is alive, and kicking as hell!

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